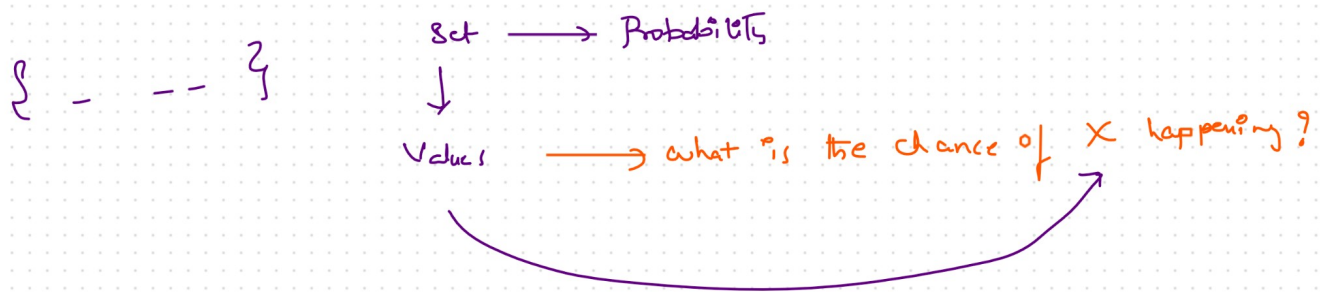


Statistics

SETS: The language of outcomes

A set is a collection of distinct items



P.1 Every possible outcome of a dice roll

Sample space (Ω) $\{1, 2, 3, 4, 5, 6\}$ (set)

event A = "roll of an even number" = $\{2, 4, 6\}$ subset

event B = "roll greater than 4" = $\{5, 6\}$ subset

event C = "roll of 9" = $\{ \}$ subset

empty set ϕ

The four set operations

$A = \{2, 4, 6\}$ $B = \{5, 6\}$ $\Omega = \{1, 2, 3, 4, 5, 6\}$

$A \cup B$ — Union

$A \cap B$ — Intersection

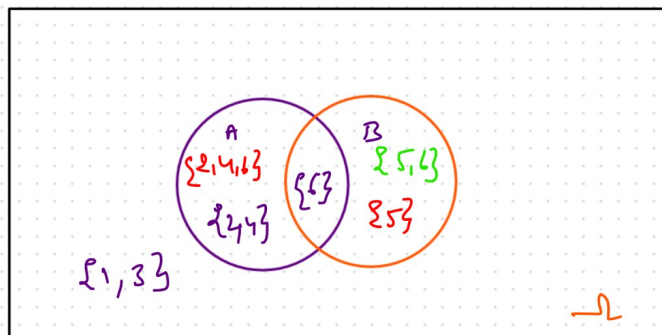
A^c — Complement

$A \setminus B \iff A - B \rightarrow$ difference

$A \cup B \rightarrow A \text{ or } B$
 $\downarrow$
 Union

everything in A, everything in B, no duplicates

$$A \cup B = A + B - A \cap B$$



$$A \cup B = A + B - A \cap B$$

$$= \{2, 4, 6\} + \{5, 6\} - \{6\}$$

$$A \cup B = \{2, 4, 5, 6\}$$

$A \cap B \rightarrow A \text{ and } B$ (values common in both sets)
 $\downarrow$ intersection

$$A \cap B = A + B - A \cup B$$

$$= \{2, 4, 6\} + \{5, 6\} - \{2, 4, 5, 6\}$$

$$= \{6\}$$

A^c (Not A)

$$= U - A$$

$$= \{1, 3, 5, 6, 8\} - \{2, 4, 6\}$$

$$= \{1, 3, 5\}$$

$A \setminus B \rightarrow A - B$

A but not B (elements in A that are not in B)

$$A - B = \{2, 4, 6\} - \{5, 6\}$$

$$= \{2, 4\}$$

Random Variable

experiment: flip 2 fair coins



1 coin $\rightarrow H, T$

2 coin $\rightarrow HH, TT, HT, TH$

function (coin flip)

$\rightarrow$ no. of heads
count

random variable
no. of numbers
numbers $\{2, 0, 1\}$

I win 10 rupees of each head

outcomes $\rightarrow$ number

$HH \rightarrow 20$

$TT \rightarrow 0$

$HT \rightarrow 10$

$TH \rightarrow 10$

Random variable $\rightarrow X$ is a function that converts outcome to a number

$$X(HH) \rightarrow 2$$

$$X(TT) \rightarrow 0$$

$$X(HT) \rightarrow 1$$

$$X(TH) \rightarrow 1$$

$\therefore X$ can take values from set: $\{0, 1, 2\}$

Probability: How likely is an event

Ex 1

Toss 1 coin

$$S = \{H, T\}$$

Event $A: \{H\}$

$$P(A) = \frac{\text{no. of favorable outcomes}}{\text{Total outcomes}}$$

$$= \frac{1}{2} = 0.5$$

So, chance the event will be Head

exple dice roll

$$S = \{1, 2, 3, 4, 5, 6\}$$

Event A : even number

$$A = \{2, 4, 6\}$$

$$P(A) = \frac{2, 4, 6}{1, 2, 3, 4, 5, 6} = \frac{3}{6} = \frac{1}{2} = 0.5$$

There is a 50% chance that if you roll a dice you will get an even number

Event (not an even number)

A → even number

not an even number — A^c

P(A^c) → not an even number

$$\sum P(A) = 1$$

$$P(A^c) = 1 - P(A)$$

$$= 1 - \frac{1}{2} = \frac{2-1}{2} = \frac{1}{2} = 0.5$$

$$P(A^c) = 0.5 \rightarrow 50\%$$

$$\frac{3}{6} + \frac{3}{6} \rightarrow 1$$

$$2, 4, 6 \quad 1, 3, 5$$

$$P(O) + P(E) = 1$$

$$S = \{H, T\}$$

$$P(H) = \frac{1}{2} \quad \frac{1}{2} + \frac{1}{2} = \frac{2}{2} = 1$$

$$P(T) = \frac{1}{2}$$



{Head faced, Tail empty}

$$P(HF) + P(HE) \rightarrow 1$$

$$P(HF) = 1 - P(HE)$$

Solution 1

$P(A) \rightarrow A \rightarrow$ not an even number

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{1, 3, 5\}$$

$$= \frac{3}{6} = \frac{1}{2} = 0.5$$

Experiment

OR condition

$$S = \{1, 2, 3, 4, 5, 6\}$$

Event

$$A : \text{even} \longrightarrow \{2, 4, 6\}$$

$$B : \text{multiple of 3} \longrightarrow \{3, 6\}$$

$P(A \cup B) \rightarrow$ probability of a number either even or multiple of 3

$$P(A) = \frac{3}{6}$$

$$P(B) = \frac{2}{6}$$

$$P(A \cap B) = \frac{1}{6}$$

$$(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{3}{6} + \frac{2}{6} - \frac{1}{6}$$

$$= \frac{5-1}{6} = \frac{4}{6} = \frac{2}{3} = 66.67\%$$

↓
Chance that number is either even or multiple of 3

Condition Probability

$$P(A|B)$$

probability of A given B already happened

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A = \text{even} = \{2, 4, 6\}$$

$$B = \text{multiple of 3} = \{3, 6\}$$

P of a even number when B happened {3, 6}

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A|B) = \frac{\frac{1}{6}}{\frac{2}{6}} = \frac{1}{2}$$

Ex: Take a king from deck & then queen card from deck

52

$$Pr(K) = \frac{4}{52}$$

$$Pr(Q) = \frac{4}{51}$$

dependent events

Toss a coin & then Tail

$$Pr(H) = \frac{1}{2} \quad Pr(T) = \frac{1}{2}$$

→ independent events

Sets → Random Variable → probabilities

discrete random variable

✓
Countable values, separate

- ↓
- 1. Heads in 2 flips
- 2. Students absent
- 3. Wickets taken

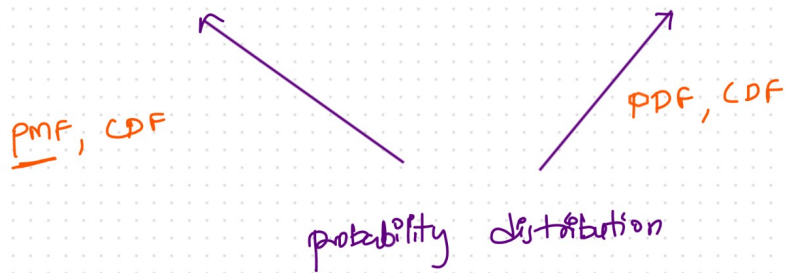
Continuous random variable

infinite possible values with in range

→ person height

→ exam start time

→ temperature



PMF : probability mass Function

→ we map probability of each & every outcome

↳ what is the probability of value x ?

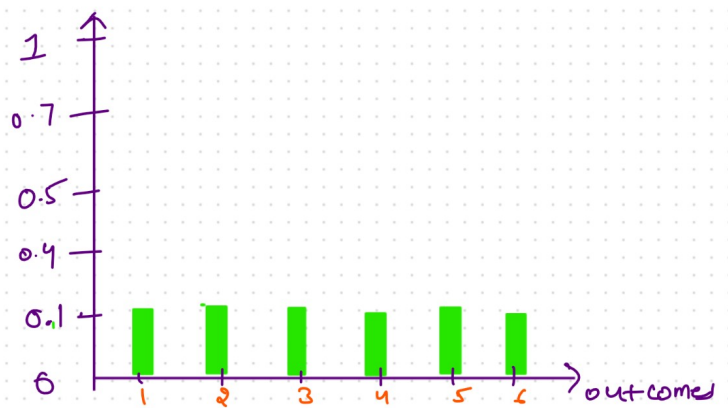
P1 roll a dice

1	—	$1/6$
2	—	$1/6$
3	—	$1/6$
4	—	$1/6$
5	—	$1/6$
6	—	$1/6$

0.167

← map

probabilities



② No. of Heads in a 2 coin flip

	X	Count
TT → 0	0	1
HT → 1	1	2
TH → 1	1	2
HH → 2	2	1

$$Pr(X=0) = \frac{1}{4} = 0.25$$

$$Pr(X=1) = \frac{2}{4} = 0.5$$

$$Pr(X=2) = \frac{1}{4} = 0.25$$

$$\sum P(x) = 1$$

$$0.25 + 0.5 + 0.25 = 1$$

Probability

